

Shathurjan Muhunthan

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Professional Summary

Second-year Aerospace Engineering student experienced in CAD, propulsion systems, robotics, and flight training. Skilled in concept-to-test workflows involving simulation, design, prototyping, manufacturing, and verification. Recognized for data-driven problem solving, precision modelling, and cross-disciplinary collaboration across aerodynamics, structures, propulsion, and controls.

Education

BEng, Aerospace Engineering *Sep 2024 – Present*

Toronto Metropolitan University

- GPA: 4.06/4.33; Dean's List (Fall 2024, Winter 2025)
- Relevant coursework: Fluid Mechanics, Thermodynamics, Materials Science, Electrical Circuits, Engineering Design & Communication

Ontario Secondary School Diploma *Feb 2023 – Jun 2024*

Westhill Collegiate Institute

Technical Competencies

Design & Modeling	SolidWorks, Onshape, CATIA; Parametric CAD; Geometry/surface design; Assembly integration; OpenRocket
Tools & Languages	ANSYS (training), MATLAB, Simulink; Arduino IDE (C/C++); C; HTML; CSS; JavaScript; Data analysis/visualization; Testing/prototyping
Professional Skills	Technical communication; Team coordination; Technical mentorship; Safety-minded testing; Event/logistics management

Engineering & Technical Experience

Hardware Team Member — Skydrone Research Project (Frontier) *Sep 2025 – Present*

- Design and analyze the physical structure of an autonomous UAV for both terrestrial and extraterrestrial use cases.
- Conduct FEA and CFD simulations using SolidWorks, ANSYS, and MATLAB/Simulink to evaluate frame stiffness and aerodynamic efficiency.
- Collaborate with electronics and software teams to align mechanical design with control and AI navigation systems.
- Support research into lightweight materials and control surface design for future autonomous deployment.

Lead-in-Training — Metropolitan Aerospace Rocket Society (MARS) *Sep 2025 – Present*

- Support student teams through the design, manufacturing, assembly, and launch of competition rockets while training under an experienced lead.
- Assist teams with preliminary rocket configuration, stability analysis, and performance evaluation using OpenRocket.
- Guide members through CAD development in Onshape, including component design, assembly integration, and design iteration.
- Assist with manufacturing, troubleshooting, launch preparation, and technical decision-making while developing team leadership skills.

Glider Pilot Trainee – Scholarship Recipient — University Soaring Society *2026 – Present*

- Awarded a flight-training scholarship to pursue a Glider Pilot Licence through the University Soaring Society.
- Progressed through supervised glider flight training and successfully completed first solo flight.
- Develop practical experience in aircraft handling, aerotow procedures, circuit operations, takeoffs, landings, and flight safety.
- Continue flight training and licensing requirements toward obtaining a Glider Pilot Licence.

Propulsion Subteam Member (Igniter Team) — TMU MetRocketry *Mar 2025 – Present*

- Conducted background research to support the team’s first spincasting approach for igniter components.
- Reviewed material options, manufacturing processes, and safety factors relevant to igniter development and spincasting.
- Attended propulsion meetings to understand igniter requirements and gain exposure to hybrid-engine fundamentals.
- Support igniter development activities while building practical knowledge of propulsion-system design, manufacturing, and testing.

Competitor – Mechanical/Structural Designer — MLP, Combat Robotics *May 2025 – Present*

- Lead mechanical design and CAD for a two-person combat robot, including frame, drivetrain, and weapon subsystems.
- Model parts in Onshape with CF-PETG and steel reinforcements to balance stiffness, impact strength, and weight.
- Manage mass budget, geometry layout, and manufacturability under beetleweight constraints.
- Integrate with electronics systems for wiring protection, component placement, and service accessibility.

Competitor-Lead Developer — NASA Space Apps Challenge Hackathon 2025 *Oct 2025*

- Designed a full-stack asteroid impact visualization platform integrating NASA NEO, GIS, and demographic data.
- Built a Node.js and Express server with front-end logic in HTML, CSS, and JavaScript using Esri ArcGIS and Open-Meteo APIs.
- Modelled crater scaling, shockwave radius, and energy propagation using open scientific datasets for hazard prediction.
- Earned NASA Space Apps “Best Storytelling” Second Runner-Up for clarity in scientific communication and system design.

Competitor-Lead CAD Designer — Metropolitan Aerospace Rocket Society (MARS) *Jan 2025 – Mar 2025*

- Modelled the aerodynamic nose cone and avionics bay for a hybrid student rocket using CATIA and MATLAB.
- Derived and implemented curvature equations to generate accurate geometry based on OpenRocket data.
- Integrated spline-based point inputs from MATLAB into CATIA for parametric surface generation.
- Ensured alignment between avionics design, structural manufacturability, and aerodynamic performance.

Embedded Systems Designer — Plant Wall Automation (STEM+ Club) *Sep 2023 – Jun 2024*

- Built an Arduino-based automated system regulating watering, lighting, humidity, and temperature for plant growth.
- Programmed C++ logic for real-time response to soil-moisture and environmental sensors.
- Wired and calibrated circuits for stable closed-loop control and reliable one-year operation.
- Collaborated with the biology team to define operational thresholds and environmental parameters.

Electronics Integrator — Autonomous Rover Project (STEM+ Club) *Sep 2023 – Jun 2024*

- Assisted in wiring, assembling, and coding a line-following autonomous rover with infrared sensors and DC motors.
- Created 3D-printed chassis parts using Bambu Studio and Arduino IDE C++ for motor control routines.
- Supported troubleshooting and debugging post-demo to identify performance inconsistencies.
- Gained foundational experience in autonomous system wiring, motion control, and sensor integration.

Leadership & Community Experience

Lead Volunteer Coordinator — Natchathiravila (Canadian Tamil Radio Annual Concert) *Jul 2023, Jul 2024, Jul 2025*

- Directed 40+ volunteers across stage operations, logistics, and technical coordination.
- Maintained safety and communication standards throughout multi-day cultural events.
- Acted as liaison between management, vendors, and volunteers.
- Recognized for organizational reliability under time-sensitive conditions.

Co-President — Wellness Warriors Club – Westhill Collegiate Institute *Feb 2023 – Jun 2024*

- Organized mental health campaigns and school-wide “Wellness Wednesdays” discussions.
- Oversaw team scheduling, outreach, and inter-grade collaboration.
- Developed communication initiatives promoting mindfulness and stress awareness.
- Strengthened leadership and coordination in a student-led environment.

Peer Tutor & STEM+ Club Member — Westhill Collegiate Institute *Feb 2023 – Jun 2024*

- Tutored over 60 hours in physics, chemistry, and mathematics, improving student academic performance.
- Produced summary sheets and practice materials adopted by peers.
- Guided small groups through hands-on STEM challenges and demonstrations.
- Supported STEM showcase logistics and presentation coaching for junior students.

Assistant Admin / Coach-in-training — Lay-Up Youth Basketball *Jul 2023 – Aug 2023*

- Managed scheduling, attendance, and inventory for youth basketball camps across multiple sites.
- Assisted coaching staff with training, gameplay supervision, and youth mentorship.
- Created tracking tools, improving resource management and participant engagement.
- Ensured coordination between coaches, parents, and program administration.

Yoga Trainee Instructor — Ananda Yogalaya Int’l, Sri Lanka *Nov 2021 – Nov 2022*

- Completed foundational training of “Yogaratna” teaching level.
- Supported mixed-level sessions emphasizing accessibility and physical safety.
- Helped organize community yoga events promoting wellness engagement.
- Strengthened composure and instructional communication in group settings.

Awards & Achievements

- University Soaring Society – Glider Flight Training Scholarship
- NASA Space Apps Hackathon 2025 – “Best Storytelling” Second Runner-Up
- Dean’s List – TMU (Fall 2024, Winter 2025)
- **Grade 12 Awards:** Chemistry (AP) Award; Scholar Award
- **Grade 11 Awards:** Chemistry (AP) Award; Physics (AP) Award; Functions (AP) Award; Scholar Award

Additional Information

Languages English (fluent), Tamil (native)